

问题 B: Gamma 刀治疗方案

立体定位放射外科, 用单一高剂量离子化射束在 X 光机精确界定下照射颅内的一个小的 3D 脑瘤, 与此同时, 并没有处方剂量的任何显著份额伤及周边的脑组织. 在这个领域中, 一般有三种形式的射束可以采用, 分别是 Gamma 刀单元, 带电重粒子射束, 以及来自直线加速器的外用高能光子束. Gamma 刀单元具备的单一高剂量离子化射束, 是 201 个钴-60 单位源通过厚重的盔状物发射出来的. 所有的 201 条射束同时交会于一个等中心(最大放射剂量点), 从而在有效剂量的水平上形成一个近似 球形的剂量分布. 照射这个等中心来达到处方剂量称为一个“shot”.

多个 shot 可以表述为不同的球. 四个可以互换的外部校准的盔状物分别具有 4, 8, 14 和 18mm 的射束通道直径, 都可以用来照射不同尺寸的体积. 对于大于一个“shot”的目标体积, 可以用多个 shot 来覆盖整个目标. 实际上, 大多数目标体积要用 1 到 15 个“shot”加以处理. 在这里, 目标体积是一个有界的通常包含数百万个点的三维数字图象. 放射外科学的目的是消除肿瘤细胞同时保存正常的结构. 由于治疗过程中会涉及物理限制和生物不确定性, 一个治疗方案就需要考虑到所有那些限制和不确定性. 一般而言, 一个最优的治疗方案需要符合如下的要求:

- ? 穿过目标体积的剂量梯度最小
- ? 为目标体积配置特异性的相同剂量轮廓线
- ? 为目标和关键器官配置特异性的剂量-体积限制条件
- ? 对正常组织或器官的整个体积照射要剂量总和最小
- ? 对指定的正常组织点的剂量要限制在忍耐剂量以下
- ? 使关键体积所需的最大剂量达到最小 在 Gamma 单元治疗方案中, 有以下限制:
- ? 禁止“shot”伸展到目标以外
- ? 禁止“shot”交迭 (避免热点)
- ? 用有效的剂量覆盖尽可能多的目标体积, 但至少 90% 目标体积要被“shot”覆盖
- ? 用尽可能少的“shot” 你的任务是用球体填充问题模型来建立最优的 Gamma 刀治疗方案, 并且提出一个求解的算法. 在设计算法时你要记住: 它必须是相当有效率的.

PROBLEM B: Gamma Knife Treatment Planning Stereotactic radiosurgery delivers a single high dose of ionizing radiation to a radiographically well-defined, small intracranial 3D brain tumor without delivering any significant fraction of the prescribed dose to the surrounding brain tissue. Three modalities are commonly used in this area; they are the gamma knife unit, heavy charged particle beams, and external high-energy photon beams from linear accelerators. The gamma knife unit delivers a single high dose of ionizing radiation emanating from 201 cobalt-60 unit sources through a heavy helmet. All 201 beams simultaneously intersect at the isocenter, resulting in a spherical (approximately) dose distribution at the effective dose levels. Irradiating the isocenter to deliver dose is termed a “shot.” Shots can be represented as different spheres. Four interchangeable outer collimator helmets with beam channel diameters of 4, 8, 14, and 18 mm are available for irradiating different size volumes. For a target volume larger than one shot, multiple shots can be used to cover the entire target. In practice, most target volumes are treated with 1 to 15 shots. The target volume is a bounded, three-dimensional digital image that usually consists of millions of points. The goal of radiosurgery is to deplete tumor cells while preserving normal structures. Since there are physical limitations and biological uncertainties involved in this therapy process, a treatment plan needs to account for all those limitations and uncertainties. In general, an optimal treatment plan is designed to meet the following requirements.

- ? Minimize the dose gradient across the target volume.
- ? Match specified isodose contours to the target volumes.

- ? Match specified dose-volume constraints of the target and critical organ.
- ? Minimize the integral dose to the entire volume of normal tissues or organs.
- ? Constrain dose to specified normal tissue points below tolerance doses.
- ? Minimize the maximum dose to critical volumes. In gamma unit treatment planning, we have the following constraints:
 - ? Prohibit shots from protruding outside the target.
 - ? Prohibit shots from overlapping (to avoid hot spots).
 - ? Cover the target volume with effective dosage as much as possible. But at least 90% of the target volume must be covered by shots.
 - ? Use as few shots as possible. Your tasks are to formulate the optimal treatment planning for a gamma knife unit as a sphere-packing problem, and propose an algorithm to find a solution. While designing your algorithm, you must keep in mind that your algorithm must be reasonably efficient.